

PROTOCOL

Aim-Core Blank Artificial Cells

FOR RESEARCH USE ONLY
STORE IN 2-8°C

Hydrogel microparticle (HMP) with biotin-binding sites on a lipid membrane surface.

Description

Aim-Core Blank Artificial Cells are non-magnetic, hydrogel-based microparticles (HMPs). They enable fast and efficient docking of biotinylated molecules, allowing customizable signal presentation to target cells. Aim-Core provides a platform to explore cell-signaling interactions with enhanced biomimicry while maintaining a simple mix-and-grow workflow.

Aim-Core allows users to attach any biotinylated signal to the particle surface for cell-like presentation. Signals bind to the artificial cells through simple mixing. This flexibility makes Aim-Core ideal for investigating niche cell subtypes, specialized workflows, or alternative species models. We offer **12 Aim-Core formulations**. Users can choose based on **degradability** under physiological conditions, **signal density** (avidin density), and **membrane fluidity**:

Catalogue Code	Degradability	Membrane Fluidity	Avidin density	Biotinylated molecule docking capacity *
AVI01-0100	Transient/ Self-degrading 7-10 days in RPMI, 37°C, 5% CO ₂	Low	Low ~ 4000 /μm ²	1.6e-17 mol/HMP
AVI02-0100			Medium ~ 14500 /μm ²	5.2e-17 mol/HMP
AVI03-0100			High ~ 44000 /μm ²	1.6e-16 mol/HMP
AVI04-0100		High	Low ~ 4000 /μm ²	1.6e-17 mol/HMP
AVI05-0100			Medium ~ 14500 /μm ²	5.2e-17 mol/HMP
AVI06-0100			High ~ 44000 /μm ²	1.6e-16 mol/HMP
AVI07-0100	Persistent/ >60 days in RPMI, 37°C, 5% CO ₂	Low	Low ~ 4000 /μm ²	1.6e-17 mol/HMP
AVI08-0100			Medium ~ 14500 /μm ²	5.2e-17 mol/HMP
AVI09-0100			High ~ 44000 /μm ²	1.6e-16 mol/HMP
AVI10-0100		High	Low ~ 4000 /μm ²	1.6e-17 mol/HMP
AVI11-0100			Medium ~ 14500 /μm ²	5.2e-17 mol/HMP
AVI12-0100			High ~ 44000 /μm ²	1.6e-16 mol/HMP
*Assume each avidin molecule has 3 free biotin binding sites				

Components

A suspension of hydrogel microparticles (HMPs) made of chemically crosslinked dextran. HMPs are 7-12 μm in diameter and coated with a phospholipid bilayer. Avidin molecules are docked on the membrane surface.

Each kit contains:

- 1 vial of Aim-Core HMPs in 0.9% NaCl at 4×10^7 particles/mL with 0.5% P/S (1.0 mL)
- 1 vial of HMP lysis buffer (100×, 1.0 mL) containing dextranase in PBS

Stability and Storage

- Shipped with ice, do not freeze, keep product refrigerated (2-8°C)
- Stable at 4°C for 24 months
- Contents are sterile when unopened

Other Required Materials

- ☐ Washing Buffer: 0.22 μm filtered, or autoclaved 1X PBS
- ☐ Docking Buffer: 0.22 μm filtered 1X PBS with 0.1% BSA, or recombinant albumin.
- ☐ Sterile centrifugation tubes
- ☐ Biotinylated molecules
- ☐ Mixing instrument

Signal Docking Protocol

Note: All steps should be performed under sterile conditions.

Pre-wash Aim-Core HMPs

- Resuspend Aim-Core HMPs by vortexing the vial to obtain a homogeneous pink suspension.
 - Note: pink colour comes from premixed dye in the lipid membrane, it indicates the membrane integrity.
 - vortex for 10 sec; avoid prolonged vortexing to prevent membrane damage.
- Aliquot desired volume of Aim-Core HMPs into microcentrifuge tubes
- Wash the aliquot once with 1 mL **Washing Buffer**. Centrifuge at $800 \times g$ for 3 minutes and decant the supernatant.

- Resuspend the pellet in **Docking Buffer**. Count HMPs number and calculate the particle concentration using a hemocytometer.

Docking of biotinylated signals

- Calculate the amount of biotinylated molecules needed for docking according to the avidin density and average biotinylated molecule docking capacity provided in the table on page 1.
- Prepare stock solutions of biotinylated molecules.
 - Calculate molar concentration of each signal you plan to dock.

$$\text{Molar concentration (mol/}\mu\text{L)} = \frac{\text{Mass conc. (mg/mL)}}{\text{Molecular weight (Da)}} \times 10^{-6}$$

For example, you have 3 types of biotinylated molecules want to dock:

Biotinylated molecule	Type	Molecular weight	Stock Conc. (mg/mL)	Molar Conc. (mol/ μ L)
A	IgG	150 kDa	0.5	3.3×10^{-12}
B	Protein	57 kDa	0.1	1.7×10^{-12}
C	Protein	30 kDa	0.1	3.3×10^{-12}

- Calculate total free biotin docking site number in Aim-Core HMP.

$$\begin{aligned} &\text{Total biotin docking site (mol)} \\ &= \text{Aim-Core HMP} \times \text{docking capacity per HMP (mol/HMP)} \end{aligned}$$

For example, for **4×10^6 Aim-Core (#AVI05-0100)**, total free biotin docking sites = $(4 \times 10^6) \times (5.2 \times 10^{-17}) = 2.1 \times 10^{-10}$ mol. **Refer to the catalogue table on page 1 for biotin docking capacity per HMP.**

- Determine Required Signal Amounts

$$\begin{aligned} &\text{Required amount of each signal (mol)} \\ &= \text{Total free biotin docking site (mol)} \times \text{Signal Ratio (\%)} \end{aligned}$$

$$\text{Volume of each signal (}\mu\text{L)} = \frac{\text{Required signal amount (mol)}}{\text{Molar cocentration (mol/}\mu\text{L)}}$$

For example, to dock 3 signals at **9%, 21% and 70% ratio**,

Biotinylated molecule	Docking ratio	Required signal (mol)	Molar Conc. (mol/ μ L)	Volume (μ L)
A	9%	1.9×10^{-11}	3.3×10^{-12}	5.7
B	21%	4.4×10^{-11}	1.7×10^{-12}	25.1
C	70%	1.5×10^{-10}	3.3×10^{-12}	44.0
Total	100%	2.1×10^{-10}	-	74.7
Docking Buffer	-	-	-	=74.7 x 2 = 149.4
Aim-Core (4×10^6 HMP)	-	-	-	100 (if 4×10^7 /mL)

3. Signal Docking Procedure

- Mix biotinylated signals in **Docking Buffer first**. Add Docking Buffer volume at 2X of the total volume of mixed biotinylated molecules.
- Add Aim-Core HMP suspension, **remember to resuspend** HMP before pipetting.
- Incubate for **20 mins at room temperature** with **gentle mixing**.
- Wash the particles 3 times with wash buffer, by centrifugation at **800x g for 3 mins**. Resuspend HMP pellet in Washing Buffer.
- Count HMP and calculate concentration using a hemocytometer.
- For storage, keep at **4°C**. Shelf life is limited by the shortest stability of molecules in the mixture.

HMP seeding

Calculate the desired HMP seeding density for each well and aliquot the suspension accordingly. Instead of applying a fixed “bead-to-cell ratio”, use the recommended HMP seeding density range of **$8 \times 10^4 \sim 1.6 \times 10^5$ HMP per cm^2** to begin with. Experienced users are encouraged to adjust seeding conditions to meet their specific experimental needs. The guiding principle is to balance the probability of interaction between HMPs and target cells while maintaining sufficient space for cell growth and expansion.

Seeding density range recommendations (Cheat sheet for busy people)

Plate	96 well	48 well	24 well	6 well	T25 flask	T75 flask
Area (cm ²)	0.32	0.95	1.9	9.5	25	75
HMP /Well	$2.5 \times 10^4 \sim 5.0 \times 10^4$	$7.6 \times 10^4 \sim 1.5 \times 10^5$	$1.5 \times 10^5 \sim 3.0 \times 10^5$	$7.6 \times 10^5 \sim 1.5 \times 10^6$	$2.0 \times 10^6 \sim 4.0 \times 10^6$	$6.0 \times 10^6 \sim 1.2 \times 10^7$

HMP cleanup

HMP will self-degrade through hydrolysis. Refer to catalogue table on page 1 for degradability of each formulation. Since the hydrolysis rate is pH-dependent, the actual degradation time may vary depending on the culture medium used.

Alternatively, HMP can be rapidly degraded enzymatically by adding HMP lysis buffer (included in the kit) directly to the culture medium at 1× final concentration (HMP lysis buffer stock concentration is 100X). After incubating at 37°C for 30 minutes, the HMP will be fully degraded. For Example, if your culture medium is 1 mL, you will add: **1 mL/100 = 10 µL** of HMP lysis buffer.

Important Note:

- ⚙ HMP lysis buffer contains dextranase, which degrades the HMP core. **Do not use this buffer if you plan to restimulate cells**, as residual enzyme will degrade newly added HMP unless thoroughly washed out.
- ⚙ It is recommended to **degrade HMP before cryopreservation**. Wash the cell pellet at least twice before cryopreservation to remove residual dextranase.

Things to note in general

When seeding cells with HMPs

Ensure cells and HMP are evenly distributed to maximize interaction.

When you see cell cluster

Signal docked HMP and target cells tend to aggregate in the well centre over time. Gently shake the culture plate to redistribute HMP and cells. **Avoid unnecessary pipette mixing in the first few days**, disturbing the HMP-cell clustering will cause suboptimal cell growth.

Website



LinkedIn



Email



AimCore AimGel

Aim-Core is part of our AimGel artificial cell line-up.

Allegrow

www.allegrowbiotech.com

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